

The proximal-mapping extension proposed in arXiv:2012.04784

Literature-status note for `fa_banach.001`

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Status. `literature_already_answered` (proximal-mapping extension only).

The original future-work direction

In its concluding paragraph, the original paper says that it would be interesting to extend its projector results to underrelaxed projectors or even proximal mappings [1, p. 25]. It separately asks for an algorithm computing the difference vector without Seeger’s algorithm. The two requests must not be conflated.

The paper cited there as initial progress is arXiv:2101.05857 [2]. Its theorem-level contents show that it carries out the proposed proximal-mapping extension, not merely a preliminary special case.

The later proximal theory

Let X be a real Hilbert space, let $f_i \in \Gamma_0(X)$ for $1 \leq i \leq m$, put $\mathbf{X} = X^m$ and $f = f_1 \oplus \cdots \oplus f_m$, and let R be the circular right shift on $\mathbf{X}$. A classical proximal cycle is a vector z satisfying

$$z = \text{Prox}_f(Rz), \quad \text{equivalently} \quad 0 \in \partial f(z) + (\text{Id} - R)z.$$

The later paper systematically studies these cycles, their gap vectors, and the fixed-point sets of the cyclic proximal compositions. In particular, when the classical gap vector $y = (y_1, \dots, y_m)$ exists, the successive fixed-point sets are translations by its coordinates.

More importantly, the theory remains meaningful when classical cycles do not exist. Write Δ for the diagonal in $\mathbf{X}$ and set

$$g = \text{cl}(f \square \iota_\Delta).$$

Theorem 4.2 (PDF pp. 13–14) proves that the generalized primal and dual problems

$$0 \in \partial g(x) + (\text{Id} - R)x, \quad x \in \Delta^\perp,$$

and

$$0 \in \partial(f^* + \iota_\Delta^*)(y) + \tfrac{1}{2}y + Ty$$

have unique solutions (where T is the skew linear operator associated with the shift in that paper). These are the generalized cycle and gap vector. Theorems 4.3 and 4.4 (PDF pp. 14–15) show precisely how this construction recovers classical proximal cycles whenever they exist.

Theorem 7.2 (PDF pp. 23–24) also supplies a forward–backward algorithm. For $0 < \gamma < 1$ and standard relaxation parameters, it iterates

$$q_n = (1 - \gamma)x_n + \gamma R x_n, \quad x_{n+1} = x_n + \lambda_n (\text{Prox}_{\gamma g} q_n - x_n).$$

It proves weak convergence to a generalized cycle and strong convergence of $Rx_n - x_n$ to the unique generalized gap vector. Thus the later paper provides structural, duality, existence, recovery, and computational results for cyclic compositions of proximal mappings at the same level at which the original paper treated projectors.

Why the algorithmic question remains outside this answer

For projectors onto $C_1, \dots, C_m$, take $f = \iota_C$ with $C = C_1 \times \dots \times C_m$. Then

$$g = \text{cl}(\iota_C \square \iota_\Delta) = \iota_{\overline{C+\Delta}}, \quad \text{Prox}_{\gamma g} = P_{\overline{C+\Delta}}.$$

Consequently, the later forward–backward scheme still calls the projector onto $\overline{C+\Delta}$. The original paper’s difficulty was exactly the lack of an explicit formula for this projector, for which it invoked Seeger’s algorithm. The proximal extension therefore does not, by itself, give the requested Seeger-free projector algorithm.

The proper conclusion is scoped: the proximal-mapping extension proposed on page 25 is answered by arXiv:2101.05857; the adjacent explicit-algorithm question is not claimed to be resolved. The original paper itself cites the later work as promising initial progress, and the present status assignment comes from checking its subsequent theorems. This packet makes no new mathematical claim.

References

- [1] S. Alwadani, H.H. Bauschke, J.P. Revalski, and X. Wang, *The difference vectors for convex sets and a resolution of the geometry conjecture*, arXiv:2012.04784, <https://arxiv.org/abs/2012.04784>.
- [2] S. Alwadani, H.H. Bauschke, and X. Wang, *Attouch–Théra Duality, Generalized Cycles and Gap Vectors*, arXiv:2101.05857, <https://arxiv.org/abs/2101.05857>.